

War on germs

In the early 1800s, many patients died in hospitals from infections picked up during surgery or childbirth. By the middle of the century, scientists had begun to understand that infections were caused by invisible germs. They campaigned to improve the cleanliness of hospitals, but faced opposition from doctors who refused to accept the new theories.

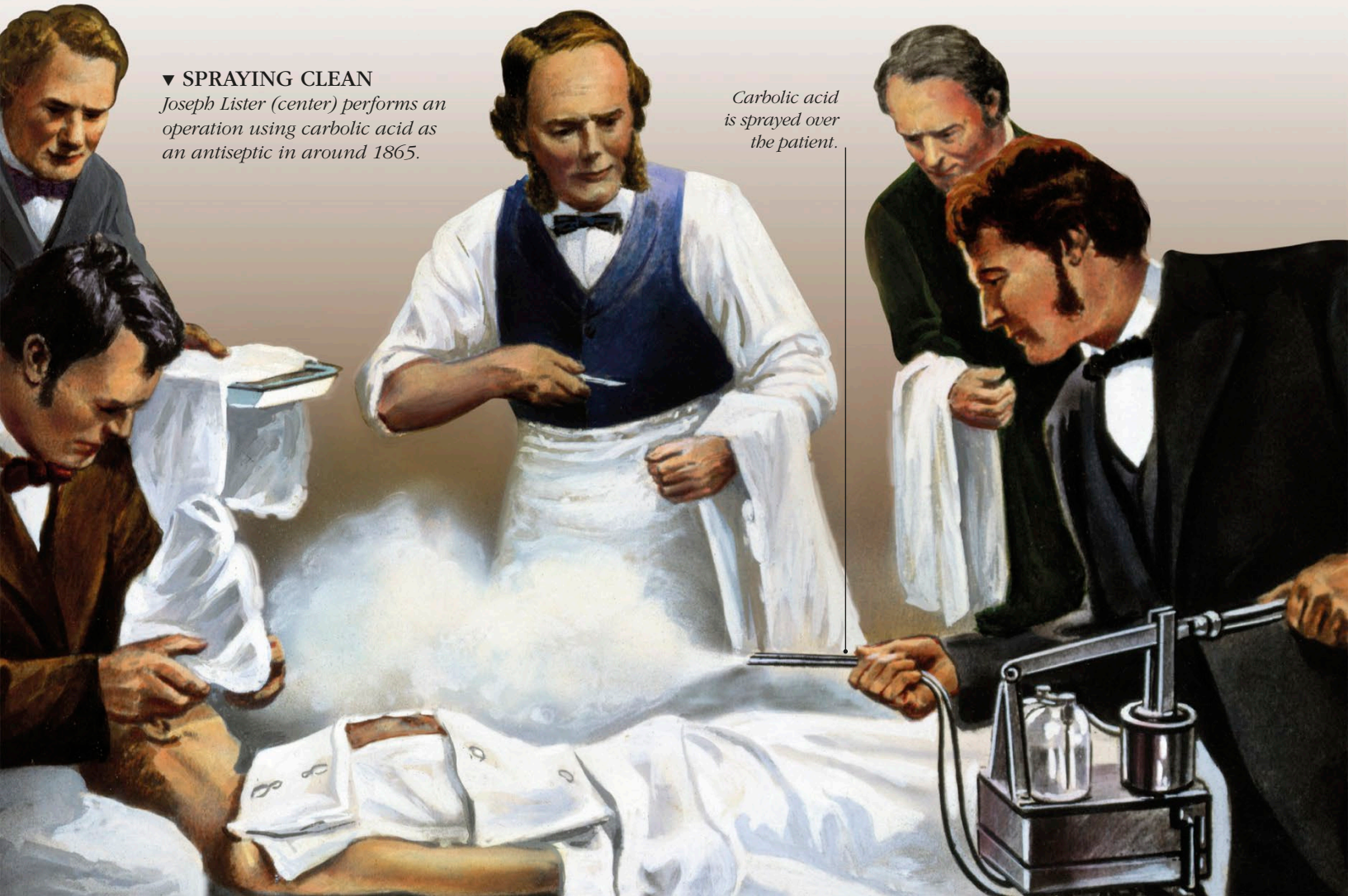
THE FIRST ANTISEPTIC

In the 1860s, the British surgeon Joseph Lister took the first practical steps to prevent airborne infections during surgery. He cleaned wounds with carbolic acid—the first antiseptic (a chemical that kills disease-causing germs). Lister also set up a machine that sprayed a fine mist of carbolic acid while he operated. This greatly reduced infection rates.

▼ SPRAYING CLEAN

Joseph Lister (center) performs an operation using carbolic acid as an antiseptic in around 1865.

Carbolic acid is sprayed over the patient.



HOSPITAL HYGIENE

Nowadays, hospital staff carefully scrub their hands before any contact with patients. The practice was pioneered by the Hungarian doctor Ignaz Semmelweis in 1847 at a hospital in Vienna, Austria. He discovered that if doctors washed their hands in mildly chlorinated water before surgery, the death rate of patients dropped.